

While Student A is building the “Ears,” **Student B** is the “**Action Specialist**.” Your job is to take the text Student A provides and make the computer actually *do* things.

Since Student A is still setting up the microphone, you can start building your logic using a keyboard first, then “plug in” the voice later.

Step 1: The “Command Center” (Setup)

Create a new Java class named `ActionEngine.java`. This class will be the “Brain” that makes decisions.

1. **Create the Method:** Write a method called `public void executeCommand(String text)`.
2. **Clean the Input:** Inside this method, make the text lowercase and remove spaces. This ensures that “Open Notepad” and “open notepad” are treated the same.

Step 2: The “Launcher” (Using `ProcessBuilder`)

Now you need to learn how Java talks to Windows. You will use a tool called `ProcessBuilder`.

1. **The “If” Logic:** Write an if statement: `if (text.contains("notepad"))`.
2. **The Launch Code:** Inside that if, write: `new ProcessBuilder("notepad.exe").start();`
3. **Try/Catch:** Java will insist you wrap this in a try-catch block in case the computer can’t find Notepad.

Step 3: Expanding the Skills (More Commands)

A professional project needs more than one trick. Add these to your `ActionEngine`:

1. **Web Search:** Use `java.awt.Desktop.getDesktop().browse(new URI("http://google.com"))`; to open the browser when you say “Open Google.”
2. **System Time:** If the user says “Time,” use `java.time.LocalDateTime.now()` to print the current time.
3. **The “Secretary” (File I/O):** This is a great OOP addition. If the user says “Write note,” create a `FileWriter` to save their text into a file named `notes.txt`.

Step 4: The “Final Assembly” (Main.java)

Once Student A is done, you will sit together and write the Main.java class. This is where you connect the two halves.

1. Create the Objects:

- o `VoiceListener ears = new VoiceListener();`
- o `ActionEngine hands = new ActionEngine();`

2. The Loop:

- o `String command = ears.listenForCommand();`
 - o `hands.executeCommand(command);`
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Summary of Student B’s Deliverables:

- ☐ A working ActionEngine class that can open at least 3 different apps/websites.
 - ☐ A “Note-taking” feature that saves text to a file (this looks very professional).
 - ☐ Error handling (so the program doesn’t crash if it hears a word it doesn’t recognize).
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How to know you are done:

If you can type “notepad” into a scanner (temporary test) and Notepad opens, your logic is perfect. When Student A finishes their part, you will simply replace your “Scanner” with their “VoiceListener.”

Quick Fairness Check:

- **Student A** handles the complex **Libraries and Hardware**.
- **Student B** handles the complex **System Logic and Commands**.

Student B, are you ready to start building the “Action” part of the team?